

HERNANDO, PANTINE B.

CHALLENGE 4 - CRYPTOGRAPHY

BSIT-3

interencdec 



Easy Cryptography picoCTF 2024 base64 browser_webshell_solvable caesar

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Description

Can you get the real meaning from this file.

Download the file [here](#).

Hints ?

1

Engaging in various decoding processes is of utmost importance

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 picoCTF{FLAG}

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This challenge is an introductory cryptography problem from **picoCTF 2024** that requires multiple stages of decoding, moving from data encoding formats to classical substitution ciphers.

Description

The challenge provides a downloadable text file and hints that "engaging in various decoding processes is of utmost importance". The initial provided string is:

```
YidkM0JxZGtwQIRYdHFhR3g2YUhsZmF6TnFIVGwzWVROclgyMHdNakV5TnpVNGZR  
PT0nCg==
```

Solution Strategy

The solution involves a three-step process: two layers of **Base64** decoding followed by a **Caesar cipher** decryption.

Step 1: First Base64 Decoding

The input string ends with `==`, a characteristic padding for Base64 encoding.

- **Input:** `YidkM0JxZGtwQIRYdHFhR3g2YUhsZmF6TnFIVGwzWVROclgyMHdNakV5TnpVNGZRPT0nCg==`
- **Tool:** `base64decode.org`
- **Output:** `b'd3BqdkpBTXtqaGx6aHlfazNqeTI3YTNrX20wMjEyNzU4fQ=='`



base64decode.org

Decode from Base64 format

Simply enter your data then push the decode button.

YidkM0JxZGtwQIRYdHFhR3g2YUhsZmF6TnFIVGwzWVROclgyMHdNakV5TnpVNGZRPT0nCg==

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8



Source character set.

☐ Decode each line separately (useful for when you have multiple entries).



Live mode OFF

Decodes in real-time as you type or paste (supports only the UTF-8 character set).



DECODE



Decodes your data into the area below.

b'd3BqdkpBTXtqaGx6aHlfazNqeTI3YTNrX20wMjEyNzU4fQ=='

Step 2: Second Base64 Decoding

The output from the first step is a Python byte string containing another Base64 encoded segment.

- **Input:** `d3BqdkpBTXtqaGx6aHlfazNqeTI3YTNrX20wMjEyNzU4fQ==`
- **Tool:** `base64decode.org`
- **Output:** `wpjvJAM{jhlzhy_k3jy9wa3k_m0212758}`

Decode from Base64 format

Simply enter your data then push the decode button.

```
'd3BqdkpBTXtqaGx6aHlfazNqeTI3YTNrX20wMjEyNzU4fQ=='
```

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8



Source character set.



Decode each line separately (useful for when you have multiple entries).



Live mode OFF

Decodes in real-time as you type or paste (supports only the UTF-8 character set).



DECODE



Decodes your data into the area below.

```
wpjvJAM{jhlzhy_k3jy9wa3k_m0212758}
```

Step 3: Caesar Cipher Decryption

The resulting string looks like a flag format but is scrambled. Using a Caesar cipher tool (like `cryptii`), we test different shifts until the "wpjvJAM" prefix becomes "picoCTF".

- **Input:** `wpjvJAM{jhlzhy_k3jy9wa3k_m0212758}`
- **Shift:** 7 positions
- **Output:** `picoCTF{caesar_d3cr9pt3d_f0212758}`

Final Result

After successfully navigating the layers of encoding and encryption, the flag is revealed:

Flag:

`picoCTF{caesar_d3cr9pt3d_f0212758}`

The screenshot shows a web browser at the URL `cryptii.com/pipes/caesar-cipher/`. The interface features three main panels: **Ciphertext**, **Caesar cipher**, and **Plaintext**. The **Ciphertext** panel contains the input `wpjvJAM{jhlzhy_k3jy9wa3k_m0212758}`. The **Caesar cipher** panel is configured with a **SHIFT** of `-7` (a→h) and an **ALPHABET** of `abcdefghijklmnopqrstuvwxyz`. The **Plaintext** panel displays the resulting output `picoCTF{caesar_d3cr9pt3d_f0212758}`. Below the cipher settings, a status bar indicates `→ Decoded 34 chars`.